



A dynamical model towards continuous single-cell cultivation using photobioreactors

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ABSTRACT

Phenomenological models for different stages of a photosynthetic system have helped predict experimental outcomes at the cell level. Similar models exist to predict process outcomes at the bioreactor scale. However, biological and process models are typically ontologically separate, as process models do not include explicit mappings of cellular processes. This work presents a model for the dynamics of photobioreactors with phenomenological reaction kinetics. The model allows for design and control integration, with insights into design parameters and operating conditions aiding both the growth of microalgae and the production of valuable lipids. The model also enables integration of process topology such that the reactor network connections yielding a complete process can be modeled alongside reaction kinetics. Therefore, we provide insights on nonlinear dependence of biomass and lipid production using microalgae cultivation, design parameters such as incident light intensity and photobioreactor radius, and process operating conditions such as flow rates.

1. Introduction

Microalgae, ranging from eukaryotic protists to prokaryotic cyanobacteria, are fast-growing photosynthetic cells found in diverse wet environments, including oceans and damp terrestrial areas, in which they play a key role as primary producers in the food chain Day et al. (1999), Li et al. (2008). Industrially, they are prized for their ability to produce essential biomolecules like lipids, proteins, and carbohydrates in bioreactor environments. They have notably been used to produce carotenoids, fatty acids, and heterologous proteins at scale Cereghino and Clegg (1999); Swartz (2001). Their resilience to saltwater environments, a trait known as halotolerance, makes them especially valuable for protein production studies in conditions where other organisms cannot thrive, and highlights their potential in developing low water footprint processes Geng et al. (2003).

In the industrial cultivation of microalgae, various autotrophic growth-supporting photobioreactors are used for both phototrophic and mixotrophic conditions. These systems are typically efficient in converting substrates into biomass but require extensive light-exposed surface areas to function optimally Wasanasathian and Peng (2007), Castañeda et al. (2023). To maximize light exposure, many photobioreactors are

constructed with narrow, elongated tubes, ensuring that the culture within is close to the light-penetrable walls. In these reactors, the culture is continuously circulated, and a stream of CO₂-enriched air flow is provided to facilitate CO₂ assimilation and O₂ production via photosynthesis Delavar and Wang (2022). Since sunlight can be variable and unreliable, especially for large or sensitive cultures, artificial lighting is often employed in pilot-scale operations, despite the associated energy costs Muller-Feuga et al. (1998).

Sufficient literature exists in developing dynamical models for single-cell cultivation at experimental level, as shall be elucidated in the following sections. These models usually range from empirical phenomenological correlations to dynamical models driven by data. The focus of such models are specific to either growth of specific system variables or optimization. However, these models often do not integrate the system dynamics with phenomenological correlations, under continuous operation, and tradeoffs in design are seldom discussed. The aim of this study is to create a dynamical model for a microalgae bioreactor using existing research and mass balances to simulate operation in varied conditions. The objectives include developing a predictive model for the bioreactor output, describing the changes in cell, lipid, and nitrogen levels over time, determining optimal operating conditions for lipid production in

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a well-mixed reactor, henceforth referred to as a continuous mixed tank reactor (MTR).

The paper is organized as follows: In Section 1, we cover the basic phenomenology critical to the growth and development of microalgae and present the model forms used to develop the kinetics of the process. Section 2 demonstrates the use of these model forms to develop the MTR process model using flow inputs and outputs. We show the results of the model in Section 3, with focus on the dependence of biomass and lipid production on the design and operating parameters, showing that the inherent nonlinearities observed between the operating conditions, and the outputs with dynamics and steady states, allows for possibility of utilization of machine learning models to estimate the final steady states. We conclude the paper in Section 4, with future directions on how to develop the dynamical model into usable phenomenology.

2. Key phenomena for microalgae cultivation

In this section, we will present the main phenomena critical to the dynamics – including cell growth, substrate uptake, product formation and the impact of light incidence on the growth. Each of the subdivided dynamics are characterized by their phenomenological relationships.

2.1. Cell growth

It is quite complex to write a unique set of biochemical reactions for cell growth in the usual form of elemental proportions of the synthesized molecules. The growth phases of microorganisms in batch cultures follow three main phases: the lag phase, the exponential growth phase and the death phase, which are all critical to a wide range of biotechnological applications (Lee (2001)).

The Monod and Droop models are standard in this domain, with the former effectively describing microbial growth under nutrient-limited conditions and the latter being more applicable to environments with higher nutrient availability.

The Monod model, a cornerstone in microbial growth kinetics, particularly under nutrient-limited conditions, articulates the relationship between microbial growth rate and the concentration of a limiting substrate. This relationship is formalized in the Monod equation drawing a parallel to the Michaelis-Menten kinetics in enzyme-catalyzed reactions (Doran (1995)):

$$\mu = \mu_{\max} \frac{S}{K_S + S} \quad (1)$$

In this equation, $\mu_{\max}$ represents the maximum specific growth rate under optimal substrate availability, while K_S , the half-saturation constant, indicates the substrate concentration at which the growth rate is half of $\mu_{\max}$. This formulation yields a hyperbolic relationship, where the growth rate initially rises with increasing substrate concentration and eventually plateaus.

The Droop Model is a cell growth model focusing on systems with unrestricted nutrient availability and the internal storage of nutrients in cells. It is predicated on the concept that cells require a minimum concentration of a specific nutrient, which they accumulate internally. This minimum nutrient level is expressed as a quota (q), calculated as the ratio of the nutrient concentration to the total biomass of the cell (X). Note that this is different from the amount of functional biomass, x . Growth rates are determined by this formula, implying that cells must reach a minimum attainable quota for growth (Droop (1968)).

$$\mu = \begin{cases} \mu_{\max}(1 - q_{\min}/q), & \text{if } q \geq q_{\min} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

The specific nutrient concentration used in the calculation varies by species and is often measured in atomic concentrations of elements like Nitrogen (N), Carbon (C), Phosphorus (P), or Silicon (Si) (Lee (2015); Sommer (1991); Wan et al. (2014)). A modified version considering,

nitrogen quota, lipid quota, amount of substrate and light incidence is considered in this article.

2.2. Substrate uptake

In microalgal cultures, specific transporters mediate the uptake of substrates such as CO₂ and nitrates, which are assimilated through photosynthesis and nitrogen metabolism pathways, respectively (Lea and Ben Miflin (2011)). CO₂ is fixed into organic molecules by RuBisCO in the Calvin-Benson-Bassham cycle, while nitrates are reduced and incorporated into amino acids. Phosphorus is integrated into adenosine triphosphate (ATP), the primary energy-carrying block of molecules, a for energy (Wang et al. (2015)). Algae cells can switch between storing excess nutrients as lipids via acyltransferase enzymes (Hu et al. (2008)) or mobilizing stored reserves when nutrients are scarce, highlighting their metabolic adaptability (Schmollinger et al. (2014)).

The substrate utilization dynamics in cellular systems are described by a set of equations that quantitatively represent the interplay between substrate consumption, product formation (in this case lipids), biomass generation, and maintenance energy requirements. The core equation for substrate consumption rate (r_S) is:

$$\begin{aligned} r_S = \frac{dS}{dt} &= - \left(\frac{1}{Y_{PS}} r_P + \frac{1}{Y_{XS}} r_x + m_S x \right) \\ &= - \left(\frac{1}{Y_{PS}} r_P + \frac{1}{Y_{XS}} \mu x + m_S x \right) \end{aligned} \quad (3)$$

In this equation, r_S signifies the rate of substrate consumption. r_P and r_x (with $r_x = \mu x$) represent the volumetric rates of product formation and biomass production, respectively. x is the functional biomass concentration, Y_{PS} and Y_{XS} are the yields of product and biomass per unit substrate, and m_S is the maintenance coefficient. The maintenance coefficient is a crucial parameter which denotes the rate at which substrate is consumed for cellular maintenance processes. Its value is species-specific and influenced by environmental conditions, including the ionic strength of the medium (Van Bodegom (2007)).

The nitrogen uptake rate in cells, crucial for growth, is governed by the external nitrogen source concentration and the internal nitrogen quota (del Rio-Chanona et al. (2017)). This uptake adheres to Michaelis-Menten kinetics, and imported nitrogen is assimilated for conversion into amino acids or nucleotides, integral to functional biomass (Lehman et al. (1975); Yoo et al. (2015); del Rio-Chanona et al. (2017)). Further refinement in modeling includes an equation proposed by Lehman et al. (Lehman et al. (1975)):

$$\rho = \rho_{\max} \left(\frac{S_2}{K_N + S_2} \right) \left(\frac{q_{\max} - q}{q_{\max} - q_{\min}} \right) \quad (4)$$

This model prevents unrealistic quota increases with higher nitrogen source concentrations, halting uptake at a maximum quota. Conversely, when near minimum quota, the uptake rate increases markedly. When the quota is well above the minimum, the growth rate results from a combination of autotrophic and heterotrophic growth rates, highlighting the complex interaction between internal quota and external nutrient supply (Adesanya et al. (2014)).

2.3. Product formation

On the other hand, product formation is driven by photosynthesis, in which CO₂ is fixed into organic compounds including glucose. These serve as substrates for lipid synthesis, in which acetyl-CoA is converted into fatty acids and then into energy-dense triacylglycerols (TAGs) under certain conditions, particularly stress Harwood and Guschina (2009). Additionally, microalgae synthesize valuable secondary metabolites such as carotenoids through the methyl-D-erythritol phosphate (MEP) pathway, with these pigments being commercially important as antioxidants (Lichtenthaler (1999)). Proteins, including enzymes and phycobiliproteins, are produced from fixed nitrogen and are used in

various biotechnological applications (Becker (2007)). For product formation, the rate r_P is detailed as:

$$r_P = \frac{dP}{dt} = Y_{Px}r_x + m_Px = Y_{Px}\mu x + m_Px = (Y_{Px}\mu + m_P)x \quad (5)$$

Here, $Y_{Px}\mu + m_P$ is the specific rate of product formation, encompassing both the direct conversion from biomass and the product formation linked to maintenance activities (Yoo et al. (2016); Yoo et al. (2015)). These are specifically related to the production using parameters Y_{Px}, m_P , linked to yield from biomass and maintenance, respectively.

Lipid synthesis in microalgae is governed by carbon fixation processes and influenced by the concentration of carbon substrates, following Michaelis-Menten kinetics. Nitrogen deprivation in microalgae has been found to enhance lipid production, with studies indicating a negative correlation between intracellular nitrogen quota and lipid synthesis rate (Abdollahi and Dubljevic (2012)). Excess intracellular lipids, however, tend to inhibit further lipid production due to the reduction in metabolic demand for lipid synthesis, as these serve as energy reserves. Various models (Yoo et al. (2016); Novak and Brune (1985)), encapsulate these dynamics, incorporating factors like maximum lipid accumulation limits. This model reflects a complex interplay between nitrogen levels, lipid content, and cell growth, where lowering nitrogen and lipid quotas may augment lipid production but can concurrently impair overall cell growth, influencing total lipid yield (Yoo et al. (2016); Novak and Brune (1985)).

$$\pi = \pi_{\max} \left(\frac{S_1}{K_L + S_1} \right) (1 - q) \left(\frac{l_{\max} - l}{l_{\max}} \right) \quad (6)$$

2.4. Light intensity

Light intensity is a critical factor for cultivating microalgae, as it directly influences their growth and metabolic processes, especially in autotrophic species that rely on photosynthesis. Hence, the focus in this work. The efficiency of photosynthesis, and thereby the production of biomass and lipids, is closely linked to the light irradiance available to the microalgae. There exists a minimum threshold of light irradiance necessary to initiate photosynthesis (Lee (2001)).

Each photosynthetic organism requires a minimum level of light, measured in watts per square meter or micromoles of photons per square meter per second, to perform photosynthesis. Below this threshold, or compensation irradiance, photosynthesis does not occur (Yamada and Sakaguchi (1982)). Beyond a certain point, however, further increases in light intensity do not enhance photosynthesis. Exceeding this optimal light intensity can lead to photoinhibition, in which photosynthetic activity is impaired due to excessive light (Sasi et al. (2011); Prioul and Chartier (1977); Yamada and Sakaguchi (1982)).

From the compensation irradiance up to the plateau, the relation between the photosynthetic activity and the irradiance is hyperbolic. In autotrophic conditions and low substrate concentration, photosynthetic activity is proportional to cell growth. The applied light intensity needs to be sufficiently high to penetrate the absorption barrier, but at the same time needs to be sufficiently low so that inhibition does not affect the growth of the cells. In steady-state conditions, it is reasonable to assume an average light intensity irradiating the sample (Bernard (2011)). If the reactor has ideal mixing, then all cells will be equally exposed to the same amount of light. In a slab geometry reactor, the average light can be calculated as Bernard et al. (Adesanya et al. (2014); Bernard (2011)) propose by using the Lambert-Beer law. In a slab reactor with depth L the average intensity can be calculated as:

$$I_{ave} = \frac{1}{L} \int_0^L I(z) dz$$

$$= \frac{1}{L} \int_0^L I_0 \exp(-\alpha B X z) dz = \frac{I_0}{\alpha B X L} (1 - \exp(-\alpha B X L)) \quad (7)$$

where α denotes the light absorption coefficient, B is the proportionality constant relating to the amount of chlorophyll in the biomass, and L denotes the mean path length.

3. Photobioreactor model description

3.1. Model of a MTR for microalgae

In this section, we present the combination of the individual phenomenon, to link them to the overall model of an idealized reactor, the MTR. The MTR model lays the groundwork for simulation of a more complex photobioreactor, envisioned as a network of MTRs. Following the development of the basic MTR model, simplifying assumptions are made, then balance equations and algebraic equations are applied to simulate the system.

The model describes the concentrations of 5 important species that are either reactants or products of the bioreactions of interest. Three of these are contained in the reactor mixture and their concentrations are calculated using the total amount of species divided by the total volume of the reactor. These include the amount of functional biomass, the amount of carbon-containing substrate, and nitrogen-containing substrate. The last 2 are measured as quotas of nitrogen and lipids, so their concentration is based on the amount that occupied the total biomass, which in turn is related to the total amount of nitrogen and lipid per reactor volume. The total biomass is then provided as the sum of functional biomass, nitrogen and lipids. All 5 are summarized in Table 1.

The total biomass (X) in the reactor hence, is a sum of the functional biomass in the reactor, the amount of nitrogen and the amount of lipids in the reactor.

3.2. Parameters and variables

Table 2 shows a list of parameters, output variables and adjustable parameters in the model. The key input parameters to the model are the flow into the MTR (F_{in}), the flow out of the MTR (F_{out}), the irradiance value (I_0) and the reactor radius (R). Apart from these, the inlet concentrations of the various species presented earlier are also given, along with their initial values. The model also presents certain adjustable parameters in the form of substrate yield (Y_{xs}), lipid yield (Y_{ls}), and maintenance cost (m_s). Other adjustable parameters are related to the specific growth constitutive equations of the different species. Considering that the rate expressions for cell growth (μ), nitrogen uptake (ρ) and lipid production (π) are already defined by the conditions of the MTR, there are 6 variables that are found because of this system of equations. These variables are the functional biomass, amount of S_1 (responsible for the growth of the biomass), amount of S_2 , amount of lipid produced, amount of N in the cell and finally the volume of the reactor. Apart from the predefined list of parameters, the yield parameters are adjusted for the biomass and lipids, for S_1 dynamics. We further, add an activator-inhibition kinetic relationship of the incident light, based on parameters, K_I, K_{I2} .

Where μ is the specific growth rate, ρ is the nitrogen uptake rate, π is the lipid production rate. Constitutive equations establish a relationship between intensive variables and variables like reaction rates or enthalpies, as detailed in (Sales-Cruz (2006)). Within the context of the microalgae MTR model, these equations specifically define the specific rates of cell growth, nitrogen absorption, and lipid generation. These rates play crucial roles in the primary balance equations and are determined based on the reactor configuration and operational setup. These constitutive equations are shown in Table 3.

In this model, total biomass is accounted as the sum of functional biomass, nitrogen quota and lipid quota: $X = x + Q + L$ Total biomass is implicitly added in the calculation of the quotas, which appear in the

Table 1
Species considered in the differential equation model.

Species	Compartment	Description	Symbol
Functional biomass	Reactor	Functionally active biomass constituting organelles, genetic material, and proteins	x
Carbon containing substrate	Reactor	CO ₂ for autotrophic metabolism, and Glucose for heterotrophic metabolism	S_1
Nitrogen-containing substrate	Reactor	Glycine	S_2
Nitrogen	Cells	Nitrogen content inside the microalgae, with the nitrogen quota given by $q = Q/X$	Q
Lipid	Cells	Lipid content inside the microalgae, with the lipid quota given by $l = L/X$	L

Table 2
Inputs, outputs and adjustable parameters.

Inputs to the ODE	$F_{in}, F_{out}, x_{in}, S_{1,in}, S_{2,in}, L_{in}, Q_{in}$
Outputs from the ODE	$x(t), S_1(t), S_2(t), L(t), Q(t), V(t)$
Adjustable parameters	$Y_{XS}, Y_{IS}, m_s, \mu_{max}, q_{min}, q_{max}, l_{min}, \rho_{max}, \pi_{max}, K_{S1}, K_N, K_L, K_I, K_{J2}, \alpha, B$

Table 3
Constitutive equations describing the process rates and corresponding variable dependencies.

Process Rate	Process Rate Equation	Variable Dependence	Condition
Cell growth	$\mu = \mu_{max} \left(1 - \frac{q_{min}}{q}\right) \left(1 - \frac{l_{min}}{l}\right) \left(\frac{S_1}{K_{S1} + S_1}\right) \left(\frac{l}{K_I + l + l^2/K_{I2}}\right)$	q, l, S_1, I	$q > q_{min}, l > l_{min}$
Nitrogen uptake	$\rho = \rho_{max} \left(\frac{S_2}{K_N + S_2}\right) \left(\frac{q_{max} - q}{q_{max} - q_{min}}\right)$	q, S_2	$q > q_{min}$
Lipid production	$\pi = \pi_{max} \left(\frac{S_1}{K_L + S_1}\right) (1 - q) \left(\frac{l_{max} - l}{l_{max}}\right)$	q, l, S_2	$q < 1, l < l_{max}$

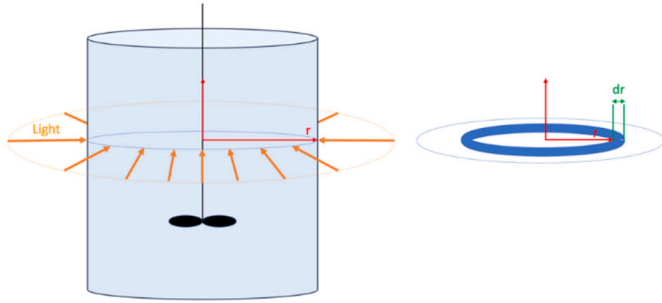


Fig. 1. Light incidence on the reactor and representation of area differential element.

three rate expressions. $q = Q/X$, $l = L/X$ are the nitrogen and lipid quota in the model. In the proposed model for the microalgae MTR, it is essential to incorporate an algebraic equation for the average light intensity experienced by the microalgae cells. Due to the light absorption by the cell culture, the irradiation intensity varies across the reactor depth. Following the Bernard et al. (Bernard (2011)) approach for a flat reactor, an average light intensity assuming constant flow can be calculated for the cylindrical geometry of the MTR. The model assumes that light is supplied evenly and perpendicularly to the reactor walls, as depicted in Fig. 1. This approximation is vital for accurately modeling the light conditions within the reactor.

The light intensity average across the whole section can be calculated using,

$$I_{ave} = \frac{1}{\pi R^2} \int_0^{A_{tot}} I(A) dA$$

$$= \frac{1}{\pi R^2} \int_0^R I_0 \exp(-\alpha BX(-r + R)) 2\pi r dr$$

$$I_{ave}(R) = \frac{2I_0}{\alpha BX} \frac{1}{R^2} \left[R - \frac{1}{\alpha BX} + \frac{\exp(-\alpha BX R)}{\alpha BX} \right]$$

The model equations are structured with all rates on the left-hand side, dependent on various other variables or rates, creating a complex web of interdependencies. For the sake of convenience, alpha and B are set numbers of 25.4804 g/g and 0.04 respectively. The model equations, such as the functional biomass rate depending on the growth rate, which in turn relies on lipid and nitrogen quotas linked to total biomass, form a loop that prevents sequential solving. Additionally, carbon and nitrogen source concentrations are affected by lipid production and nitrogen uptake rates, which are also dependent on these concentrations, creating a situation where all related equations must be solved simultaneously due to their autocatalytic nature.

3.3. Overall photobioreactor mixed tank model

Mass balances on the reactor are described for the 5 species in Eq. (REFS). Each balance yields a differential equation that describes the evolution of the concentration of each species over time. The parameters in these mass balance equations are driven by 3 constitutive equations dependent on substrates, irradiance level, and specific nitrogen and lipid quotas in the cell. These are equations below:

$$\frac{dx}{dt} = \mu x + \frac{F_{in}}{V} (x_{in} - x) \quad (8)$$

$$\frac{dS_1}{dt} = -\frac{1}{Y_{XS}} \mu x - \frac{1}{Y_{IS}} \pi x - m_s x + \frac{F_{in}}{V} (S_{1,in} - S_1) \quad (9)$$

$$\frac{dS_2}{dt} = -\rho x + \frac{F_{in}}{V} (S_{2,in} - S_2) \quad (10)$$

$$\frac{dQ}{dt} = \rho x - \mu Q + \frac{F_{in}}{V} (Q_{in} - Q) \quad (11)$$

$$\frac{dL}{dt} = \pi x - \mu L + \frac{F_{in}}{V} (L_{in} - L) \quad (12)$$

$$\frac{dV}{dt} = F_{in} - F_{out} \quad (13)$$

Table 4Model parameters for *Chlorella protothecoides* and *Chlorella vulgaris* from literature.

Parameter	Value	Units	Ref
Y_{xs}	0.9597	g/g	(Yoo et al. (2016))
Y_{ls}	0.1908	g/g	(Yoo et al. (2016))
m_s	0.0167	h-1	(Mirzaie et al. (2016))
$\mu_{\max}$	0.0418	h-1	(Yoo et al. (2016))
$q_{\min}$	0.0196	g/g	(Yoo et al. (2016))
$q_{\max}$	0.2109	g/g	(Yoo et al. (2016))
$l_{\min}$	0.0006	g/g	(Yoo et al. (2016))
$l_{\max}$	0.6995	g/g	(Yoo et al. (2016))
$\rho_{\max}$	0.1197	h-1	(Yoo et al. (2016))
$\pi_{\max}$	0.0762	h-1	(Yoo et al. (2016))
K_{S1}	0.1002	g/L	(Yoo et al. (2016))
K_N	0.5793	g/L	(Yoo et al. (2016))
K_L	12.5596	g/L	(Yoo et al. (2016))
K_I	66.5337	$\mu\text{mol m}^{-2} \text{s}^{-1}$	(Yoo et al. (2016))
K_{I2}	100.0	$\mu\text{mol m}^{-2} \text{s}^{-1}$	(del Rio-Chanona et al. (2018), Lee et al. (1987))

For the sake of simplicity, we consider the case where $F_{\text{out}} = F_{\text{in}}$, resulting in a constant volume. Parameters μ, ρ, π are given by,

$$\mu = \mu_{\max} \left(1 - \frac{q_{\min}}{q}\right) \left(1 - \frac{l_{\min}}{l}\right) \left(\frac{S_1}{K_{S1} + S_1}\right) \left(\frac{I}{K_I + I + I^2/K_{I2}}\right) \quad (14)$$

$$\rho = \rho_{\max} \left(\frac{S_2}{K_N + S_2}\right) \left(\frac{q_{\max} - q}{q_{\max} - q_{\min}}\right) \quad (15)$$

$$\pi = \pi_{\max} \left(\frac{S_1}{K_L + S_1}\right) (1 - q) \left(\frac{l_{\max} - l}{l_{\max}}\right) \quad (16)$$

We see that these equations require variables related to the nitrogen quota (q) and lipid quota (l)

$$X = x + Q + L \quad (17)$$

$$q = Q/X \quad (18)$$

$$l = L/X \quad (19)$$

$$I_{\text{ave}}(R) = \frac{2I_0}{\alpha BX} \frac{1}{R^2} \left[R - \frac{1}{\alpha BX} + \frac{\exp(-\alpha BX R)}{\alpha BX} \right] \quad (20)$$

3.4. Input parameters

The MTR model has been tested with a simple case where the inlet flow contains a specific concentration of the species of interest and the reaction takes place inside. The MTR has a total volume of 270 L. The adjustable parameters of the model have been extracted from literature and shown in Table 4.

The other input parameters that describe the configurations of the reactor are stated in Table 5 and 6.

A dynamic solution of the MTR model was achieved by simulating the changes in species concentrations over time using the 4th order Runge-Kutta method. Initially, species concentrations were set to zero, with all species being introduced through the inlet flow. A critical factor influencing the reactor behavior is the rate and composition of this inlet flow, which dictates the rate at which the reactor is charged with substrates or biomass. To calculate the concentrations of species at the steady state, the rates in the model equations are equated to zero and the remaining system of algebraic equations was solved.

We now have a system of dynamical algebraic equations that can be solved to estimate the evolution of substrates and biomass in a continuous stirred tank reactor. The functional dependence of different variables is shown in the supplementary material. The system was simulated in Python, and the functional implementation can also be seen in the supplementary material. The next section we will elucidate the results of these models for varying operating inputs of the model.

Table 5

Base-case parameters and initial values used for the differential algebraic model.

Parameters	Value	Units	
I_0	200	$\mu\text{mol m}^{-2} \text{s}^{-1}$	Arbitrary, within common range (Adesanya et al. (2014); Ibarz and Augusto (2015); Turon et al. (2015))
V	270	L	Total volume of a photobioreactor
R	15	cm	Arbitrary
x_{in}	0	g/L	No cells inlet to the reactor. Inoculation takes place first before the reactor start up and inlet only contains substrates
$S_{1,\text{in}}$	60	g/L	Arbitrary, Below solubility
$S_{2,\text{in}}$	20	g/L	Arbitrary, Below solubility
Q_{in}	0	g/L	Same as x_{in}
L_{in}	0	g/L	Same as x_{in}
α	25.4804	$\text{m}^2 \text{g}^{-1}$	(Adesanya et al. (2014))
B	0.04		(Adesanya et al. (2014); Paniagua-Michel (2015))

Table 6

Initial values used for the differential algebraic model.

Variable	Value	Units	
$x(0)$	0.25	g/L	Initial concentration after inoculation
$S1(0)$	0	g/L	No substrate initially in the reactor
$S2(0)$	0	g/L	No substrate initially in the reactor
$Q(0)$	0.025	g/L	10% of functional biomass, Arbitrary within common range (Panis and Carreon (2016))
$L(0)$	0.025	g/L	10% of the functional biomass within reported ranges (Udayan et al. (2023))

3.5. Experimental setup

To conduct the present study, we utilized the *Chlorella sorokiniana* strain. For optimal growth, we employed BG-11 as the cultivation medium, which consists of NaNO_3 (17.6 mM), K_2HPO_4 (0.23 mM), $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (0.3 mM), $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ (0.24 mM), Citric Acid $\cdot \text{H}_2\text{O}$ (0.031 mM), Ferric Ammonium Citrate (0.021 mM), $\text{Na}_2\text{EDTA} \cdot 2\text{H}_2\text{O}$ (0.0027 mM), Na_2CO_3 (0.19 mM), and trace elements. The experimental setups for the 100 ml and 3 L cultures involved a light intensity of 80 PPFD, with an 8/16 light/dark cycle, and an air injection rate of 5 L/min.

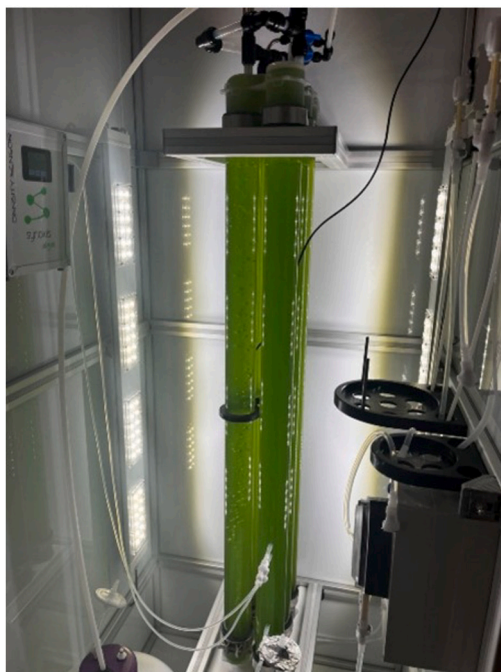


Fig. 2. LUCY Photo-bioreactor.

During the large-scale cultivation phase, we utilized the Synoxis Algae 20-liter photo-bioreactor, referred to as “LUCY.” This system features Spiral Airlift Tubular (SALT) technology, along with an advanced control system. The closed-circuit design of the photobioreactor is intended to prevent the intrusion of contaminants and to maintain the consistency of the culture. The chamber is regulated by a controller that facilitates the monitoring and automation of the culture’s conditions. This system is capable of tracking various parameters, including light intensity, pH, temperature, air and CO₂ injections, medium replenishment, sampling, optical density, dissolved oxygen content, and pH adjustments through the use of CO₂ (Fig. 2).

In the 20 L photobioreactor, we maintained the same culture media while adjusting the conditions to a light intensity of $275 \mu\text{mol m}^{-2} \text{s}^{-1}$ and an airflow rate of 0.4 liters per minute for both air inputs. The pH was set to 7, and the temperature was kept at 28 °C. We continuously monitored pH, optical density (OD), dissolved oxygen (DO), temperature, and light intensity using real-time sensors. Dry weight and cell count were measured using a standard curve developed in the laboratory, and daily samples were collected to verify morphology and check for contaminants.

4. Results and discussion

To validate the dynamics of the model, the process model was solved for different inflow conditions, keeping the outflow equal to the inlet flow rate. The system is varied for different design and operating criteria. The evolution of the concentrations of the important species related to biomass production is shown in Fig. 3, for varying inlet flow rates. The simulation was run for 1250 hours. While this high simulation time of 3 months is not practical as cells may not be viable for such long duration, the simulation time is considered to identify the steady state values of the system.

The performance of the reactor at steady state is significantly influenced by the inlet flow rate, as demonstrated in Fig. 3. Notably, at lower flow rates, higher conversion is achieved, and more biomass produced due to increased average residence time. Raising the inlet flow rate initially boosts biomass production in the bioreactor by providing more substrate, which sustains longer periods of cell growth. In contrast, at higher flow rates, there is less time for biomass to consume sub-

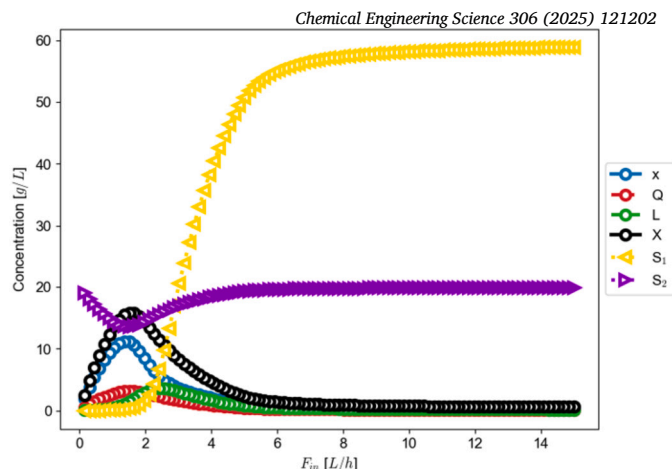
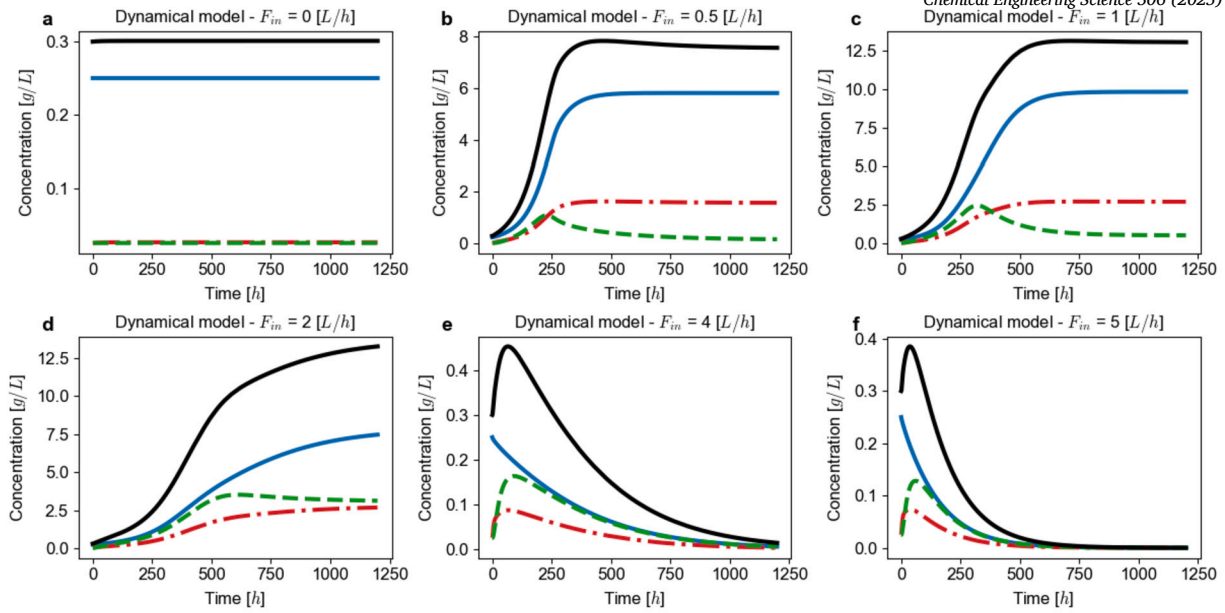
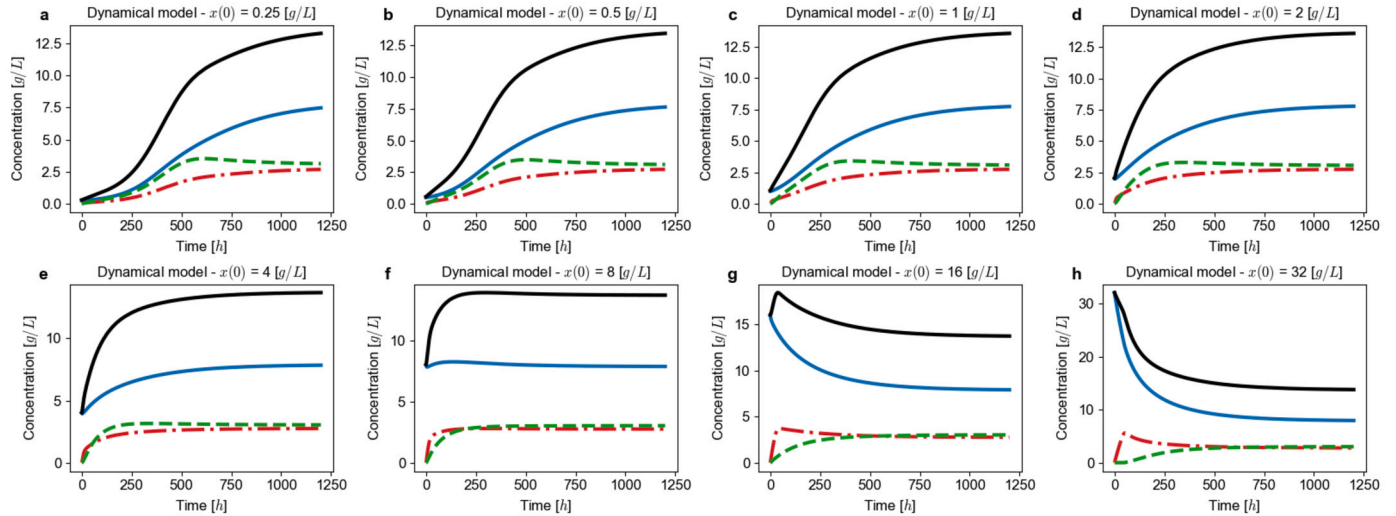


Fig. 3. Final concentrations of functional biomass, nitrogen, lipid and total biomass, with substrates S_1 and S_2 , for the MTR model at different inlet flows.

strates, resulting in lower biomass concentration and higher substrate concentrations, S_1 and S_2 , in the output. This observation is crucial for industrial applications in the continuous manufacturing setting, where a decision must be made between prioritizing high outlet concentration or rapid attainment of steady state and maximal production. The amount of consumption of the substrates by the biomass for its growths can also be understood mechanistically, using these steady state values, and dynamical simulations shown later. Biomass constituents like x , Q , and L follow similar trend of dependence on the inlet flow rate. Over certain ranges in the inlet flow values, as biomass concentration is at its peak (~ 2 L/h) we also see a high value of the lipid concentrations. However, post this flow rate eventually, lipid levels also fall when substrate availability for the biomass wanes due to washout. This is also seen in the steady-state values of the substrate concentration where it reaches its inlet concentrations at high-flow rates.

We can observe the same distribution in the dynamical simulation in Fig. 4. The figure shows that there is a critical flow rate beyond which further increases result in washout—a condition where the outlet flow rate is so high that it exceeds the rate at which biomass can grow, causing a decline in biomass concentration over time, validating our model. At moderate inlet flows of 1.0 L/h and 2.0 L/h, biomass production is more efficient, with 2.0 L/h yielding the highest productivity before reaching the tipping point of washout. Therefore, to optimize biomass yield, inlet flow rates should be managed to avoid surpassing this critical washout limit. It is seen that this washout phenomenology is captured with the developed model. However, the simulation suggests that reaching a steady state and peak production takes too long for practical industrial application, implying that the conditions tested may not be ideal for large-scale microalgae cultivation. The tradeoff between the time to achieve steady state vs. amount of biomass and lipids needs to be investigated further.

Usually, biomass concentration drops because growth, respiration, and product formation rates cannot keep up with the outlet flowrate. To fix this, one could either increase the starting biomass concentration or add cells into the inlet flow to ensure more cells are available for growth, thus preventing washout. To elucidate this, the dependence on the initial concentration of the biomass was run with an inlet flow (F_{in}) of 2 L/h, corresponding to maximum biomass production. In Fig. 5, a higher initial biomass concentration boosts maximum productivity, suggesting that the initial biomass concentrations having a higher value benefits production. This observation through the model is also observed in practice in industry, where the inoculum is typically cultivated to larger concentrations to ensure higher productivity. It is of value where such heuristic understanding might be coupled with model-based understanding as well. The figure, also shows that adding biomass to the inlet flow allows the system to reach a steady state faster and signifi-

Fig. 4. Simulation of the dynamic evolution of the MTR with different inflow rates F_{in} .Fig. 5. Simulation of the dynamic evolution of the MTR with different initial functional biomass concentrations x_0 , with $F_{in} = 2.0$ L/h.

cantly increases the outlet biomass concentration, suggesting a viable method for industrial applications to sustain cell growth.

Other important parameters of the model, critical for design and operation of the photobioreactor are the reactor radius (R) and light intensity (I_0). The impact of these parameters on biomass concentration are presented in Fig. 6 and Fig. 7. While Fig. 6 shows the dynamic evolution of biomass concentration (Fig. 6a, c) and the lipid concentration (Fig. 6b, d), Fig. 7 shows the values at the end of the simulation time after 3200 hours of simulation. Based on the trajectories in Fig. 6, these may be considered as steady state values for the operating conditions.

Lower radii of reactors are more conducive to higher biomass production. This pronounced effect in smaller reactors is due to shorter residence times, which necessitate higher light intensity for cells to absorb sufficient light, consume substrates, and produce lipids efficiently. While biomass amounts are lower in the dynamics, there is a “phase transition”-like effect in the lipid production rate. Lower radii do not aid in lipid production. However, there is a peak in production of lipids (Fig. 7b), with increasing radius of the reactor. This highlights the fact that increasing biomass and lipid production could be achieved by changing the design configurations of the reactor. We also observe

this dependence with varying light intensity (I_0) (Fig. 7c,d). Low light conditions do not significantly impact productivity but going above a threshold leads to photoinhibition, reducing productivity. As flow rates increase, the effect of light intensity becomes more critical, with higher rates necessitating a precise balance between light exposure and flow to optimize cell growth and avoid photoinhibition.

Fig. 7 shows how changes in reactor radius and light intensity specifically influence biomass production, indicating that optimal lipid yields require different conditions than those for maximum biomass growth. In fact, lipid production shows distinct trends when compared to total biomass. A reactor radius of 2 meters ($V/R = 135$ m²) results in the highest lipid productivity, indicating an optimal balance where increased cell growth boosts lipid output but excessive growth does not dilute lipid content within cells. High light intensity reduces lipid quotas due to photoinhibition, while lower intensities favor lipid production. The best outcomes occur at an intensity of $50 \mu\text{mol m}^{-2} \text{s}^{-1}$, at which lipid concentration peaks are sustained over longer periods. We note that this dependence of the final biomass concentration and lipid concentration does affect the design of the reactor, as under high biomass

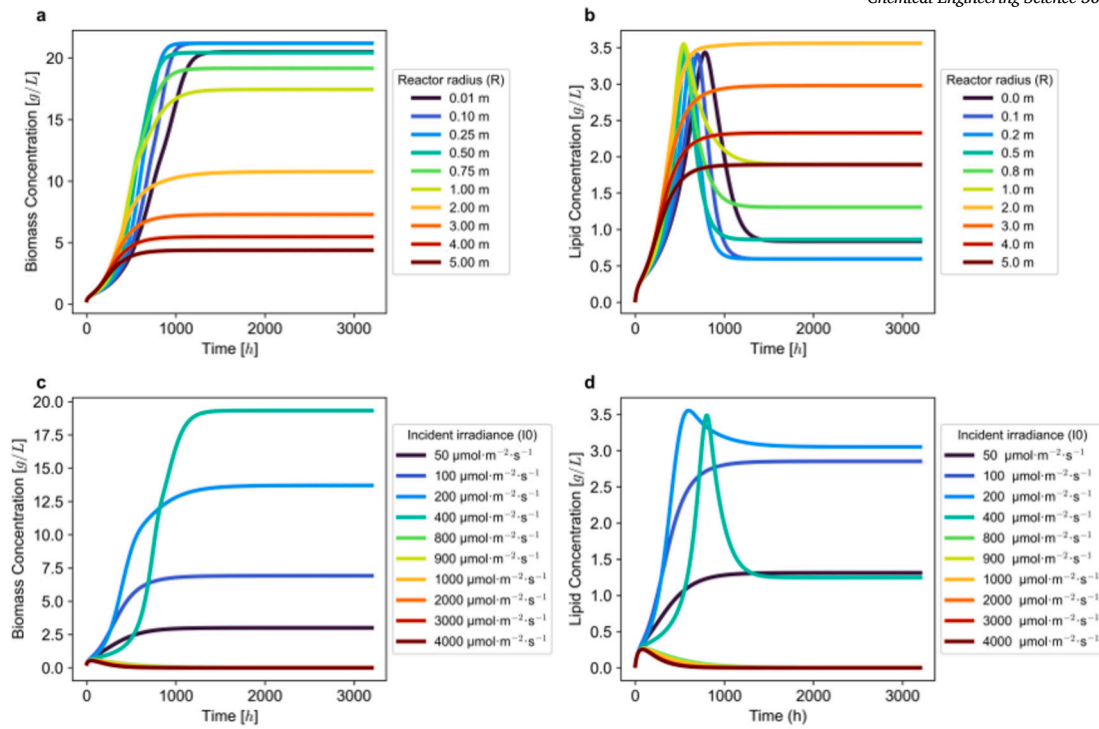


Fig. 6. Dynamic evolution of the total biomass and lipid production in a MTR with different (a, b) radius and (c,d) incident light irradiance.

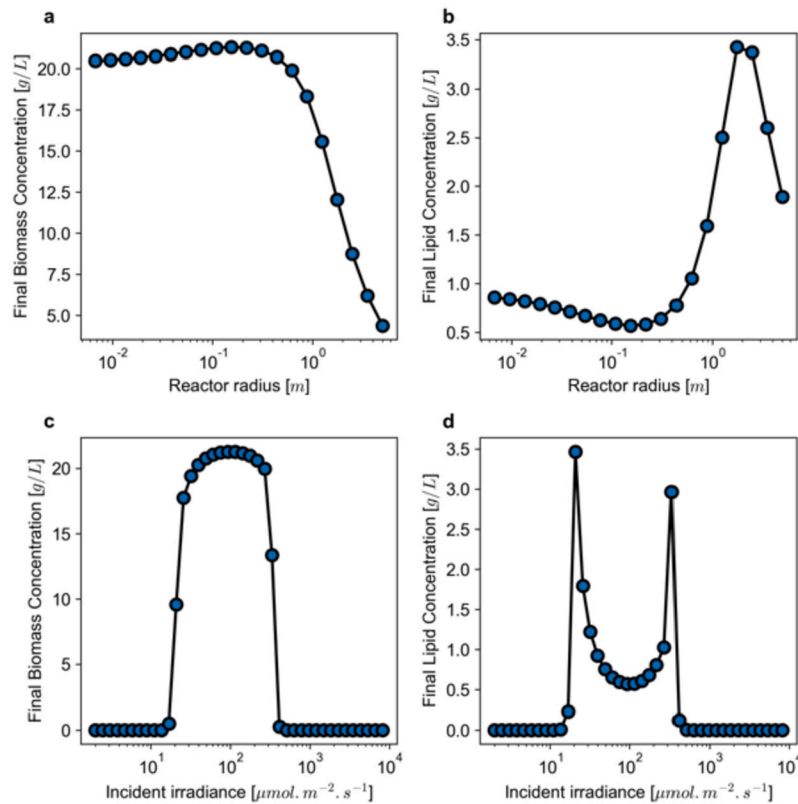


Fig. 7. Final biomass concentration and lipid concentration after 3200 hours of simulation for different MTR (a, b) radius (c, d) incident light irradiance.

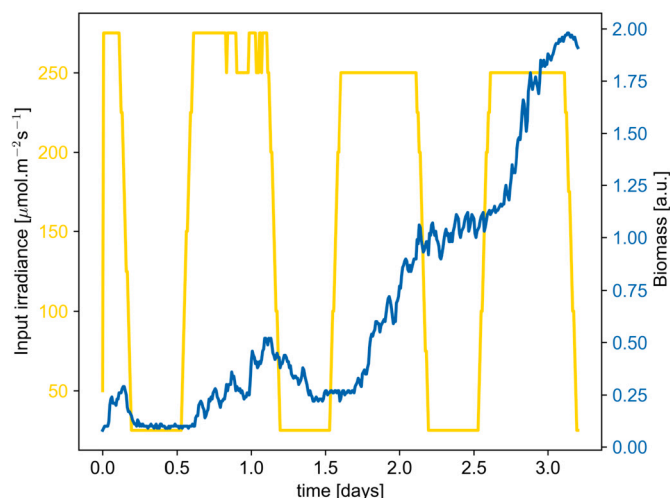


Fig. 8. Raw data from the 20 L photobioreactor.

concentrations, there is a possibility of very low irradiance, hence affecting the growth of the biomass.

4.1. Validation of insights from the model using photobioreactor data

As mentioned previously, data of incident light and biomass amount (correlated using the optical density) is collected from a 20 L photobioreactor (Fig. 8). It is of note that the model cannot be used to uniquely identify the parameter set resulting in the trajectories shown. However, we simulate our MTR model with zero inlet and similar conditions as earlier, with varying light intensity (light/dark cycles of 16/8 hours) for 20 days, to see if a qualitative capture in the dynamics may be possible. We observe varying trends in functional biomass, nitrogen quota and lipid quota, in the light and dark cycles (Fig. 9).

From the photobioreactor data, we see that in the early stages, high incident light (light phase) shows a positive growth rate of the biomass which decays in the dark phase. Note here that the light phase corresponds to optimal condition of growth of the biomass (Fig. 7c), which in turn corresponds to a minimum in lipid production (Fig. 7d). A shift into the dark phase changes the growth rate of the microalgae, as would be the case using the model. Note that our model shows the negative specific growth rate μ only when $q < q_{\min}$ that is the nitrogen quota is lower than the minimum required quota or corresponding the lipid quota, $l < l_{\min}$. The raw data would suggest that this is initially the case in the dark phase. However, this is not the case in our model simulation (Fig. 9). The growth rate of the functional biomass does change from the light phase to the dark phase, however the overall biomass follows a sigmoidal growth curve, typical of various population growth model. The final biomass concentration reaches a peak of 11.1 g/L. This reduction in the biomass from the optimal value of 20 g/L at constant irradiance of 200 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (Fig. 7c), is also accompanied by an increase in lipid concentration from a minimum of 0.8 g/L (Fig. 7d) to 3.9 g/L. This suggests that biomass and lipid concentrations can be varied at the end of the process operation by changing the light-dark cycle times.

This insight is also relevant in case of optimization of the process and control. For example, to reduce energy costs and improve productivity, it is clear that amount of light supplied to the photobioreactor is essential. For the same geometry of the photobioreactor, changing the light/dark cycle can potentially reduce the amount of energy usage and at the same time increase the productivity of the cells towards lipids.

We note here that, since biomass is estimated using optical density sensors, there are concerns that optical density measurements for microalgae may provide inaccurate estimates of actual biomass Solovchenko (2023). The changes in growth rate in certain periods of the dark phases during the experiment also suggest that other factors

like temperature or pH may also be affecting the system, apart from the nitrogen/lipid quota and the incident light. We also note that the extended simulation time may be a challenge for industrial applications. However, there are avenues to lower these using some of the insights garnered using our dynamical system model dependence, and changing the operating conditions and design parameters accordingly to suit the application.

5. Conclusions

In this work, a dynamical model for culture dynamics is presented. Reactor dynamics are modeled using a MTR formulation, using phenomenological models. The mathematical model captures the culture dynamics both qualitatively and quantitatively. The integration of various parameters, including flow rates, light intensity, and reactor geometry, coupled with differential and algebraic equations, enables the prediction of biomass and nutrient concentrations over time. The work can be extended to a specific photobioreactor using series approximations of MTRs, the accompanying model for tubular photobioreactors (PBRs) offers a realistic approach to simulating a chain of connected MTRs, reflecting the characteristics of the Jumbo® PBR, thus offering insights into the practical application of these systems in algal biotechnology.

The robustness of the MTR model is evidenced by its capacity to simulate diverse operational conditions, as revealed in the case studies. These studies highlight the delicate balance between inlet flow rates and residence times, influencing steady state and efficiency. A smaller reactor radius, while beneficial for biomass productivity due to higher light availability, may not similarly enhance lipid production. Moreover, the extrapolation of these findings into practical application warrants the integration of real-world data, especially concerning the species-specific parameters that currently rely on literature values. Experimentation with different microalgae species and bioreactor configurations would allow for a more tailored model calibration, ensuring its validity across different strains and environmental conditions. The observed nonlinearity and phase transitions in process outputs for biomass and lipids production, also allows for top-down simplification, integration of design and control schemes, and the use of machine learning for process characterization and simulation of metabolic models under-the-hood for process output estimation.

Despite its limitations in modeling variability within reactors, the model and the observed dynamics pave the way for future improvements and research. Addressing the temperature and pH dependency of kinetic rates, incorporating the effects of CO_2 dissolution on culture pH, and fitting model parameters to experimental data for a range of microalgae species could refine the model further. An important avenue would be to use approaches like extended Kalman Filter, for both denoising the data and model parameter prediction. Incorporating spatial variations in the reactor as additional features to the model, instead of taking a computationally harder route, would also help in generating fast and efficient solution methodologies with little experimentation. Such enhancements would expand model applicability, potentially making it a more versatile tool for predicting bioreactor performance under a wider array of conditions.

In conclusion, the models present a sophisticated and highly functional approach to understanding and optimizing microalgae cultivation in MTR systems. While it is of note that the model is idealised, various systems of photobioreactors, including tubular reactors, can be modeled as a series of such stirred tank reactors. Having such simple models aids in achieving fast solutions, and the approach can be extended to other bioprocesses such as single-cell protein production, where the design and operational inputs have considerable flexibility and uncertainty, respectively. The model equations are amenable to fine tuning for new species and their specific growth curves. The path forward includes refining these models with experimental data, broadening their scope, and incorporating additional physical parameters. This iterative process of

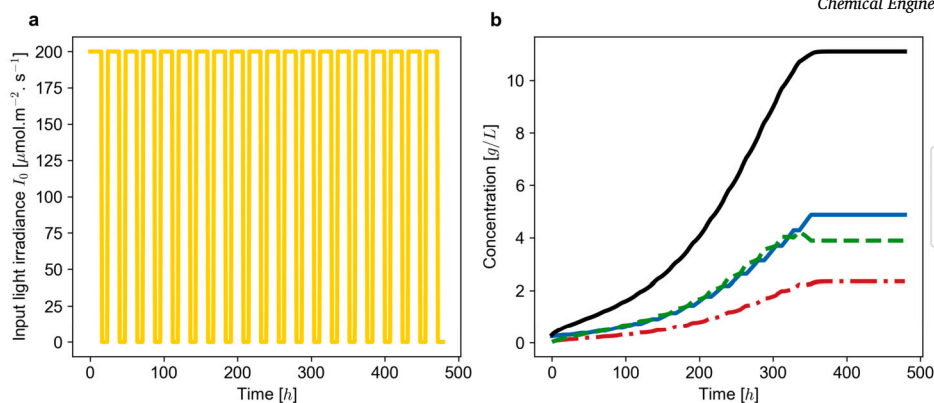


Fig. 9. Sample model simulation for a 20 L photobioreactor with varying light intensity.

model enhancement will likely yield significant benefits for the field of bioprocessing and algal biotechnology.

CRediT authorship contribution statement

Abhishek Sivaram: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Alireza Mehrdadfar:** Writing – review & editing, Formal analysis, Data curation. **Christian Euler:** Writing – review & editing, Methodology, Investigation. **Sayed Soheil Mansouri:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sayed Soheil Mansouri reports financial support was provided by Technical University of Denmark. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ces.2025.121202>.

Data availability

Data will be made available on request.

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